Frozen Phase Holonomy for a Graph-Directed Hénon Hardy–Fredholm Owner

Route-A structural certificate C129

Abstract

We freeze a fifth-root character of the integer branch translations in a strongly separated graph-directed affine Hénon system. The resulting weighted-composition operator is trace class on a direct sum of bidisc Hardy spaces. We prove its all-order trace law, Fredholm lattice product, and primitive-cycle expansion. Two assignments with the same unordered image centers and identical untwisted determinant have different twisted determinants, while the trivial character recovers the prior owner exactly. This is finite-character position sensitivity, not geometric reconstruction or a target-facing statement.

1 Frozen system and primitive cycles

On three copies of $\mathbb{D}_3^2$, set

$$A = \begin{pmatrix} 3/16 & -1/32 \\ 1/4 & 0 \end{pmatrix}, \quad \phi_j(z) = Az + (t_j, 0), \quad t = (-2, 0, 2),$$

and freeze

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad c = (1/2, 1/3, 1/5), \quad \chi(m) = \zeta_5^m,$$

where ζ_5 is primitive. The eigenvalues of A are $1/8, 1/16$. The two coordinate image radii are $21/32$ and $3/4$; the largest first-coordinate extent is $85/32 < 3$, and adjacent images have gap $2 - 2(21/32) = 11/16 > 0$. Thus the branches are compactly contained and pairwise disjoint.

Every admissible cyclic word w has affine linear part $A^{|w|}$ and one fixed point. Strong separation makes its itinerary unique, so primitive necklaces biject with primitive geometric cycles at all periods. The exact finite replay is

period	1	2	3	4	5	6	7	8
rooted closed words	1	3	7	11	21	39	71	131
primitive cycles	1	1	2	2	4	5	10	15

For example, 012 has phase points

$$\tfrac{1}{19929}(38912, -1600) \to \tfrac{1}{19929}(-32512, 9728) \to \tfrac{1}{19929}(-6400, -8128),$$

monodromy A^3, rational weight $1/30$, and holonomy $\zeta_5^5 = 1$.

2 Twisted Hardy owner

Put $W_\chi = B\,\mathrm{diag}(c_j \chi(t_j))$ and define

$$\mathcal{H} = \bigoplus_{i=0}^{2} H^2(\mathbb{D}_3^2), \qquad (\mathcal{L}_\chi f)_i(z) = \sum_j B_{ij} c_j \chi(t_j) f_j(\phi_j(z)).$$

All branch images lie in a concentric bidisc with restriction ratio $\rho = 85/96 < 1$. Total degree m has multiplicity $m+1$, so the compact-interior factorization has summable majorant $\sum_{m \ge 0}(m+1)\rho^m$. Hence $\mathcal{L}_\chi$ is trace class. Translations lower degree and leave the diagonal graded blocks unchanged. Consequently, for every $n \ge 1$,

$$\operatorname{Tr} \mathcal{L}_\chi^n = \frac{\operatorname{Tr} W_\chi^n}{(1 - 8^{-n})(1 - 16^{-n})}. \tag{1}$$

The numerator is the exact phase-weighted sum over rooted closed words. Trace class and (1) give the normally convergent entire product

$$D_\chi(z) = \det(I - z\mathcal{L}_\chi) = \prod_{r,s\geq 0} \det(I - z8^{-r}16^{-s}W_\chi). \tag{2}$$

Indeed $\sum_{r,s} 8^{-r}16^{-s} < \infty$. Grouping rooted words by primitive cycle and repetition yields

$$\log D_\chi(z) = -\sum_{[\gamma]}\sum_{m\geq 1} \frac{(c_\gamma \chi(M_\gamma) z^{\ell_\gamma})^m}{m\det(I - A^{m\ell_\gamma})},$$

where M_γ is the translation sum around the word. These formulas are all-period; period eight is only a replay prefix.

3 Exact controls and progress

To keep the two character tests logically separate, compute first in $\mathbb{Q}[\mathbb{Z}/5]$ with basis $e_0, \ldots, e_4$ and $e_a e_b = e_{a+b \bmod 5}$. Primitive-character evaluation sends $e_k \mapsto \zeta_5^k$, whereas augmentation sends every $e_k \mapsto 1$. Thus trivial degeneration is not obtained by the invalid substitution $\zeta_5 = 1$ inside the cyclotomic quotient. The source symbolic determinant is

$$\det(I - zW_\chi) = 1 - \tfrac{1}{2}\zeta_5^3 z - \tfrac{1}{6}\zeta_5^3 z^2 - \tfrac{1}{30}z^3. \tag{3}$$

Now assign the same unordered centers to branches as $t' = (0, -2, 2)$, leaving A, B, c fixed. Strong separation and every untwisted trace and determinant remain identical, but

$$\det(I - zW'_\chi) = 1 - \tfrac{1}{2}z - \tfrac{1}{6}\zeta_5^3 z^2 - \tfrac{1}{30}z^3. \tag{4}$$

The linear coefficients of the full Hardy determinants are therefore $-(64/105)\zeta_5^3$ and $-64/105$. At the trivial character the phases in (3)–(4) become one, so both operators recover the C124 trace and determinant at every order. The control 012 phase points also move exactly to

$$\tfrac{1}{19929}(32512, -9728), \quad \tfrac{1}{19929}(6400, 8128), \quad \tfrac{1}{19929}(-38912, 1600).$$

Progress over the prior gate. C124's entire determinant was blind to all branch translations. C129 retains its same all-period cycle owner and global trace-class space but distinguishes the displayed translation assignment through a frozen phase. The character sees only residues modulo five and branch assignment, so it cannot recover complete geometry.

4 Validation and Route-A boundary

The exact producer, independent checker (52 assertions), SymPy reconstruction (49 checks, finite sections through dimension 30), byte replay, and 23 repaired-hash hostile mutations pass. No external source or tolerance enters.

The strict tuple is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`), overall `ROUTE_A_EXPLORATORY`. The frozen $U(1)$ character is a formal phase lift, not a natural unitary, Hamiltonian, or metaplectic quantization. No target divisor or target global structure is compared. We claim no prime-like correspondence, arithmetic/local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Riemann-zero relation. Route B is unauthorized; the literal firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.